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Battery pack, battery rack comprising the same and energy storage system

Abstract

A battery pack includes at least one cell assembly including a plurality of battery cells, an energy consumer configured to discharge the at least one cell assembly when thermal runaway occurs in at least one of the plurality of battery cells, and a storage unit in which the energy consumer is received inside, and configured to discharge at least part of the energy consumer to outside when the energy consumer discharges the cell assembly.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5227259	12/1992	Weaver et al.	N/A	N/A
9385399	12/2015	Ryu	N/A	H01M 10/42
2003/0001578	12/2002	Lam	324/426	H01H 13/70
2010/0026242	12/2009	Forsl�w	N/A	N/A
2013/0224574	12/2012	Favaretto	N/A	N/A
2013/0328392	12/2012	Desbois- Renaudin et al.	N/A	N/A
2015/0207133	12/2014	Holl et al.	N/A	N/A
2016/0114695	12/2015	Holgers et al.	N/A	N/A
2017/0133872	12/2016	Masuda	N/A	H01M 10/4257
2018/0198294	12/2017	Sheeks	N/A	H02J 7/007194
2019/0157636	12/2018	Miler	N/A	H01M 50/30
2019/0299799	12/2018	Hinterberger et al.	N/A	N/A
2020/0036047	12/2019	Aikens et al.	N/A	N/A
2020/0036050	12/2019	Shimura et al.	N/A	N/A
2021/0226290	12/2020	Lee	N/A	H01M 10/6566
2021/0296718	12/2020	Li	N/A	H02J 7/00309

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
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3054014	12/2017	CA	H01M 10/4257
1047603987	12/2014	CN	N/A
108299178	12/2016	CN	N/A
207818832	12/2017	CN	N/A
2010-515496	12/2009	JP	N/A
5722251	12/2014	JP	N/A
2018-524695	12/2017	JP	N/A
6440116	12/2017	JP	N/A
2020-17517	12/2019	JP	N/A
10-2019-0071455	12/2018	KR	N/A
10-2019-0077650	12/2018	KR	N/A
10-2019-0135077	12/2018	KR	N/A
WO 2018/186496	12/2015	WO	N/A
WO 2018/129516	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report (PCT/ISA/210) issued in PCT/KR2021/005569, dated Aug. 19, 2021.
cited by applicant
Extended European Search Report for European Application No. 21807849.1, dated Feb. 26, 2024.
cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a battery pack, a battery rack comprising the same and an energy storage system, and more particularly, to a battery pack with improved safety against thermal runaway of a plurality of battery cells.

(2) The present application claims the benefit of Korean Patent Application No. 10-2020-0061807 filed on May 22, 2020 with the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND ART

(3) Currently, commercially available secondary batteries include nickel cadmium batteries, nickel hydride batteries, nickel zinc batteries, lithium secondary batteries, etc., and among them, lithium secondary batteries have little or no memory effect, and thus they are gaining more attention than nickel-based secondary batteries for their advantages that recharging can be done whenever it is convenient, the self-discharge rate is very low and the energy density is high.

(4) The lithium secondary battery mainly uses lithium-based oxide and a carbon material for a positive electrode active material and a negative electrode active material respectively. The lithium secondary battery includes an electrode assembly including a positive electrode plate coated with the positive electrode active material, a negative electrode plate coated with the negative electrode active material and a separator interposed between, and a hermetically sealed packaging material or

battery pouch case in which the electrode assembly is received together with an electrolyte solution.

(5) More recently, secondary batteries are being widely used in not only small devices such as portable electronic products but also medium- and large-scale devices such as vehicles and energy storage systems (ESSs). For use in medium- and large-scale devices, many secondary batteries are electrically connected to increase the capacity and output. In particular, pouch-type secondary batteries are widely used in medium- and large-scale devices because they are easy to stack.

(6) With the growing need for a large-capacity structure for use as an energy storage source, there is an increasing demand for a battery pack including a plurality of secondary batteries electrically connected in series and/or parallel, a battery module to receive the secondary batteries and a battery management system (BMS).

(7) However, the battery pack or rack includes a plurality of battery modules, and when a fire or explosion occurs due to thermal runaway in any of the plurality of secondary batteries of each battery module, heat or flames may spread to adjacent secondary batteries, causing secondary fires or explosions, and accordingly, many efforts have been made to prevent secondary fires or explosions.

(8) Accordingly, when thermal runaway occurs in any secondary battery of the battery pack or the battery rack, thermal runaway of the battery pack has been mitigated by cooling the battery pack or discharging stored electrical energy.

(9) However, in the process of discharging the electrical energy stored in the battery pack, resistance heat of a resistor that consumes the electrical energy may cause damage to the internal components of the battery pack or increase the temperature of the plurality of secondary batteries, causing thermal runaway of the secondary battery or impeding the cooling of the plurality of secondary batteries.

DISCLOSURE

Technical Problem

(10) The present disclosure is designed to solve the above-described problem, and therefore the present disclosure is directed to providing a battery pack with improved safety against thermal runaway of a plurality of battery cells.

(11) These and other objects and advantages of the present disclosure can be understood by the following description, and will be apparent from the embodiments of the present disclosure. In addition, it will be readily appreciated that the objects and advantages of the present disclosure can be realized by means and combinations thereof.

Technical Solution

(12) To achieve the above-described object, a battery pack according to the present disclosure includes at least one cell assembly including a plurality of battery cells, an energy consumer configured to discharge the at least one cell assembly when thermal runaway occurs in at least one of the plurality of battery cells, and a storage unit in which the energy consumer is received inside, and configured to discharge at least part of the energy consumer to outside when the energy consumer discharges the at least one cell assembly.

(13) Additionally, the storage unit may include a discharger configured to support the energy consumer in a direction opposite to a discharge direction of the energy consumer, and the discharger may be configured to melt by heat generated by the energy consumer to discharge at least part of the energy consumer to the outside.

(14) Furthermore, the storage unit may include a discharge door which is closed to support in a direction opposite to a discharge direction of the energy consumer, and the discharge door may be configured to be released from a closed state and be opened when the thermal runaway occurs.

(15) Additionally, the battery pack may further include a cooler including a refrigerant to cool the energy consumer discharged to the outside.

(16) Further, the energy consumer may include a switch, and a resistor which is electrically

connected to the at least one cell assembly by operation of the switch and configured to consume electrical energy stored in the at least one cell assembly.

(17) Additionally, the switch may be configured to be turned on by a physical change made by the heat generated from the at least one cell assembly.

(18) Furthermore, the battery pack may further include a sensor to sense the thermal runaway in the plurality of battery cells, and a control unit to output a control signal for turning on the switch to electrically connect the at least one cell assembly to the resistor when the control unit receives a signal notifying the thermal runaway from the sensor.

(19) Additionally, the at least one cell assembly, the resistor and the switch may be interconnected by an electrical line, and the electrical line may have a coil shape.

(20) Furthermore, the at least one cell assembly may include at least two cell assemblies, and the energy consumer may be configured to discharge a first cell assembly including the battery cell in which the thermal runaway occurred among the at least two cell assemblies or a second cell assembly among the at least two cell assemblies adjacent to the first cell assembly.

(21) Further, to achieve the above-described object, a battery rack according to the present disclosure includes a battery pack and a rack case in which the battery pack is received.

(22) Additionally, to achieve the above-described object, an energy storage system according to the present disclosure includes at least one battery rack.

Advantageous Effects

(23) According to an aspect of the present disclosure, the battery pack of the present disclosure includes the energy consumption unit and the storage unit, thereby preventing propagation of thermal runaway to the adjacent other battery cell or the other cell assembly when thermal runaway occurs in any of the plurality of battery cells. Further, the present disclosure may discharge the energy consumption unit out of the storage unit, thereby preventing the temperature rise of the component of the battery pack, for example, the other cell assembly caused by the resistance heat generated from the energy consumption unit. Further, the energy consumption unit discharged out of the storage unit may be cooled quickly when meeting the external air, thereby preventing damage to an external device by the heat of the energy consumption unit.

(24) Additionally, according to an aspect of an embodiment of the present disclosure, the battery pack of the present disclosure includes the discharge unit configured to melt away by the heat generated by the energy consumption unit to discharge at least part of the energy consumption unit to the outside, thereby inducing the energy consumption unit to be spontaneously discharged to the outside using the resistance heat generated from the energy consumption unit without a separate discharge device. Accordingly, due to the simple configuration, it is possible to save the manufacturing cost and effectively reduce the malfunction likelihood.

(25) Further, according to another aspect of the present disclosure, the battery pack of the present disclosure includes the coil shaped electrical line, thereby reducing impacts caused by a fall of the resistor using the elastic force of the coil shape when the resistor is discharged out of the storage unit. Accordingly, it is possible to prevent the coil shaped electrical line of the present disclosure from being cut when a force of the falling resistor is transmitted to the electrical line.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings illustrate preferred embodiments of the present disclosure and, together with the detailed disclosure, serve to provide further understanding of the technical spirit of the present disclosure. However, the present disclosure is not to be construed as being limited to the drawings.

(2) FIG. 1 is a schematic perspective view of a battery pack according to an embodiment of the

present disclosure.

(3) FIG. 2 is a schematic perspective view of cell assemblies of a battery pack according to an embodiment of the present disclosure.

(4) FIG. 3 is a schematic conceptual diagram showing components of a battery pack according to an embodiment of the present disclosure.

(5) FIG. 4 is a schematic cross-sectional view of the battery pack of FIG. 1, taken along the line C-C'.

(6) FIG. 5 is a schematic cross-sectional view showing the internal configuration of components of a battery pack according to an embodiment of the present disclosure.

(7) FIG. 6 is a schematic cross-sectional view showing the internal configuration of components of a battery pack according to another embodiment of the present disclosure.

(8) FIG. 7 is a schematic cross-sectional view showing the internal configuration of components of a battery pack according to another embodiment of the present disclosure.

(9) FIG. 8 is a schematic perspective view of a battery rack according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) Hereinafter, the preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms or words used in the specification and the appended claims should not be construed as being limited to general and dictionary meanings, but rather interpreted based on the meanings and concepts corresponding to the technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define the terms appropriately for the best explanation.

(11) Therefore, the embodiments described herein and illustrations shown in the drawings are just a most preferred embodiment of the present disclosure, but not intended to fully describe the technical aspects of the present disclosure, so it should be understood that a variety of other equivalents and modifications could have been made thereto at the time that the application was filed.

(12) FIG. 1 is a schematic perspective view of a battery pack according to an embodiment of the present disclosure. FIG. 2 is a schematic perspective view of cell assemblies of the battery pack according to an embodiment of the present disclosure. Additionally, FIG. 3 is a schematic conceptual diagram showing the components of the battery pack according to an embodiment of the present disclosure. In FIG. 1, the X axis direction is the right direction, the Y axis direction is the rear direction, and the Z axis direction is the up direction.

(13) Referring to FIGS. 1 to 3, the battery pack **200** according to the present disclosure may include at least one cell assembly **210** including a plurality of battery cells **211**, an energy consumption unit **220** and a storage unit **227**.

(14) Specifically, the battery cell **211** may be a pouch-type battery cell **211**. For example, as shown in FIG. 2, each of three cell assemblies **210** may include **21** pouch-type battery cells **211** stacked side by side in the front-rear direction (y direction).

(15) In particular, the pouch-type battery cell **211** may include an electrode assembly (not shown), an electrolyte solution (not shown) and a pouch.

(16) Further, as shown in FIG. 2, the battery cell **211** may have an electrode terminal **211a** at two opposite ends (in the X direction) with respect to the center of the battery cell **211**. For example, as shown in FIG. 2, each battery cell **211** of the cell assembly **210** may have a positive electrode terminal **211a** and a negative electrode terminal (not shown) extending in the horizontal direction (X direction).

(17) However, the battery pack **200** according to the present disclosure is not limited to the above-described pouch-type battery cell **211** and may employ various types of battery cells **211** known at the time of filing the patent application.

(18) Additionally, the battery pack **200** may include at least one busbar **121** configured to

electrically connect the plurality of battery cells **211** and at least two busbar frames **120** configured to mount the at least one busbar **121** on the outer side. The at least two busbar frames **120** may be provided on the left and right sides (X direction) of the cell assembly **210** respectively.

(19) Specifically, the busbar **121** may include a conductive metal, for example, copper, aluminum and nickel.

(20) Additionally, the busbar frame **120** may include an electrically insulating material. For example, the busbar frame **120** may include a plastic material. More specifically, the plastic material may be polyvinyl chloride.

(21) Meanwhile, the battery pack **200** may include a pack housing **250**. The pack housing **250** may have an internal space in which the at least one cell assembly **210** is received. For example, the pack housing **250** may accommodate three cell assemblies **210**.

(22) Meanwhile, when thermal runaway occurs in at least one of the plurality of battery cells **211**, the energy consumption unit **220** may be configured to discharge the at least one cell assembly **210**. That is, the energy consumption unit **220** may be electrically connected to a power terminal **214** (a positive power terminal and a negative power terminal) of the cell assembly **210** to intentionally cause an electrical short circuit. Accordingly, the power of the cell assembly **210** electrically connected to the energy consumption unit **220** may be consumed quickly. In other words, the energy consumption unit **220** may be configured to consume electrical energy of the cell assembly **210** quickly.

(23) Here, the 'thermal runaway' refers to a positive feedback that the temperature rise of any of the plurality of battery cells **211** increases the energy release rate of the battery cell **211** and the adjacent battery cell **211**, and in turn, this will accelerate the temperature rise. For example, thermal runaway may occur at 230° C. or above, and the battery cell **211** may spontaneously ignite.

(24) For example, the energy consumption unit **220** may be configured to apply power to the resistor **223**, a power resistor or an electric motor to consume the power of the battery pack **200**.

(25) Additionally, the energy consumption unit **220** may include an electrical short circuit. The electrical short circuit may be a circuit electrically connected between the resistor **223** of the energy consumption unit **220** and the cell assembly **210**. That is, both the external positive electrode terminal and the external negative electrode terminal of the cell assembly **210** may be electrically connected to the resistor **223** to cause an intentional electrical short circuit.

(26) Further, the storage unit **227** may be configured to receive the energy consumption unit **220** inside. The storage unit **227** may have, for example, a rectangular box shaped frame. The energy consumption unit **220** may be positioned in the internal space of the rectangular box shaped frame **235**. The storage unit **227** may be connected to the outer wall of the pack housing **250**.

(27) For example, the storage unit **227** may have an open part so that the storage unit **227** is in communication with the inside of the pack housing **250**. However, the storage unit **227** is not limited thereto, and may not be in communication with the inside of the pack housing **250**. That is, the storage unit **227** may have a space that is isolated from the outside.

(28) Additionally, when the energy consumption unit **220** discharges the cell assembly **210**, the storage unit **227** may be configured to discharge at least part of the energy consumption unit **220** to the outside. That is, when the energy consumption unit **220** discharges the cell assembly **210**, a part of the storage unit **227** may be open so that the energy consumption unit **220** moves to the outside.

(29) According to this configuration of the present disclosure, the present disclosure includes the energy consumption unit **220** and the storage unit **227**, thereby preventing propagation of thermal runaway to the adjacent other battery cell **211** or the other cell assembly **210** when thermal runaway occurs in any of the plurality of battery cells **211**. Further, the present disclosure may discharge the energy consumption unit **220** out of the storage unit **227**, thereby preventing the temperature rise of the component of the battery pack **200**, for example, the other cell assembly **210** caused by the resistance heat generated from the energy consumption unit **220**. Further, the energy consumption unit **220** discharged out of the storage unit **227** may be cooled quickly when meeting the external

air, thereby preventing damage to an external device by the heat of the energy consumption unit **220**.

(30) FIG. 4 is a schematic cross-sectional view showing the internal configuration of the components of the battery pack according to an embodiment of the present disclosure. Additionally, FIG. 5 is a schematic cross-sectional view showing the internal configuration of the components of the battery pack according to an embodiment of the present disclosure.

(31) Referring to FIGS. 4 and 5 together with FIG. 2, the storage unit **227** may have one open side through which the storage unit **227** is in communication with the inside of the pack housing **250**. An electrical line **221** may be provided to establish an electrical connection between the cell assembly **210** and the resistor **223** through the open side of the storage unit **227**.

(32) Further, the storage unit **227** may include a discharge unit **227a**. The discharge unit **227a** may be configured to support the energy consumption unit **220** in a direction opposite to the discharge direction of the energy consumption unit **220**. For example, as shown in FIG. 4, the storage unit **227** may include the discharge unit **227a** configured to support the bottom of the energy consumption unit **220** in the upward direction. That is, the discharge unit **227a** may be configured to support the energy consumption unit **220** in a direction opposite to the gravitational direction to place the energy consumption unit **220** in the storage unit **227**.

(33) Additionally, the discharge unit **227a** may be configured to melt away by the heat generated by the energy consumption unit **220** to discharge at least part of the energy consumption unit **220** to the outside. For example, the discharge unit **227a** may include a material **227a1** that melts at the temperature between about 150° C. and 300° C. The material may be, for example, paraffin or metal. The metal may be, for example, tin or lead.

(34) According to this configuration of the present disclosure, the present disclosure includes the discharge unit **227a** configured to melt away by the heat generated by the energy consumption unit **220** to discharge at least part of the energy consumption unit **220** to the outside, thereby inducing the energy consumption unit **220** to be spontaneously discharged to the outside using the resistance heat generated from the energy consumption unit **220** without a separate discharge device. Accordingly, due to the simple configuration, it is possible to save the manufacturing cost and effectively reduce the malfunction likelihood.

(35) FIG. 6 is a schematic cross-sectional view showing the internal configuration of the components of a battery pack according to another embodiment of the present disclosure. Additionally, FIG. 7 is a schematic cross-sectional view showing the internal configuration of the components of the battery pack according to another embodiment of the present disclosure.

(36) Referring to FIGS. 6 and 7 together with FIG. 3, the storage unit **227** of the energy consumption unit **220A** according to another embodiment of the present disclosure may include a discharge door **227b1**. The discharge door **227b1** may be configured to support in a direction opposite to the discharge direction of the resistor **223** of the energy consumption unit **220A**. The discharge door **227b1** may be closed to prevent the energy consumption unit **220A** from being discharged. The discharge door **227b1** may be configured to be released from the closed state and be opened when thermal runaway occurs.

(37) For example, the storage unit **227** may include a stop **227c** and a hinge member **227d**. The storage unit **227** may be configured to connect one side of the discharge door **227b1** of the hinge member **227d** to the outer wall of the storage unit **227**. That is, the discharge door **227b1** may be configured to rotatably move by the hinge member **227d**. The stop **227c** may be configured to keep the other side of the discharge door **227b1** from moving in the downward direction (gravitational direction).

(38) That is, the stop **227c** may be in the shape of a bar that extends in one direction. The stop **227c** may be disposed on the outer wall (the lower wall) of the storage unit **227**. The stop **227c** may be configured to move in the extended lengthwise direction. The stop **227c** may be configured to keep the discharge door **227b1** in the closed state or allow the discharge door **227b1** to rotatably move

according to the location in the lengthwise direction.

(39) Referring back to FIGS. 4 and 5, the battery pack **200** of the present disclosure may further include a cooling unit **240** configured to cool the energy consumption unit **220** discharged to the outside. For example, the cooling unit **240** may include a water tank **243** and a refrigerant **241** received in the water tank **243**. The refrigerant **241** may be, for example, an insulating oil having an electrically insulating property and a high specific heat. When the energy consumption unit **220** is discharged from the storage unit **227**, the cooling unit **240** may be configured to immerse the energy consumption unit **220** in the refrigerant **241** received in the water tank **243**.

(40) According to this configuration of the present disclosure, the present disclosure further includes the cooling unit **240**, thereby safely storing the energy consumption unit **220** discharged from the storage unit **227**. That is, since the energy consumption unit **220** may generate high temperature by consuming the power of the cell assembly **210**, and when contacting with an external device or the user, may be damaged, the present disclosure may quickly cool and safely store the energy consumption unit **220** using the cooling unit **240**.

(41) Referring back to FIGS. 1 to 4, the energy consumption unit **220** of the present disclosure may include an electrical line **221**, a resistor **223** and a switch **225**.

(42) Specifically, the electrical line **221** may be electrically connected to the power terminal **214** of the at least one cell assembly **210**. The power terminal **214** may be a negative power terminal or a positive power terminal. The cell assembly **210**, the resistor **223** and the switch **225** may be interconnected by the electrical line **221**. The electrical line **221** may include a metal wire and an electrically insulating coating that covers the wire.

(43) Additionally, the resistor **223** may be electrically connected to the electrical line **221** and configured to consume the stored electrical energy of the cell assembly **210**. For example, the resistor **223** may include carbon, a metal or an oxidized metal having a predetermined specific electrical resistance or above. The metal may be, for example, alloys of copper, aluminum and nickel.

(44) Further, the resistor **223** may be electrically connected to the positive electrode terminal and the negative electrode terminal of the cell assembly **210** to cause an electrical short circuit. The resistor **223** may be configured to convert the power of the cell assembly **210** into resistance heat.

(45) Additionally, the switch **225** may be configured to control the electrical connection between the at least one cell assembly **210** and the resistor **223** by the on-off operation. The switch **225** may be provided on the electrical line **221**. For example, as shown in FIG. 1, two switches **225** may be provided on two electrical lines **221** respectively. When comparing with inclusion of one switch **225**, in case that the two switches **225** are provided, it is possible to prevent unnecessary power consumption of the cell assembly **210** caused by malfunction of any of the switches **225**. The switch **225** may be configured to control electrical connection between the cell assembly **210** and the resistor **223**.

(46) For example, the switch **225** is in an off state, the switch **225** may be open to disallow the electricity of the electrical line **221** to flow, and when the switch **225** is in an on state, the switch **225** may be closed to allow the electricity of the electrical line **221** to flow.

(47) Further, the switch **225** may be configured to be turned on by a physical change made by the influence of the heat generated from the cell assembly **210**. For example, the switch **225** may be a temperature switch which opens or closes a contact point in response to the temperature using displacement resulting from expansion and contraction of a temperature sensing body such as bimetal with the temperature change of the cell assembly **210**.

(48) According to this configuration of the present disclosure, the present disclosure includes the switch **225** which opens or closes the contact point in response to the temperature, so the switch **225** may operate using the temperature change of the cell assembly **210** without a separate device for controlling the switch **225**. Accordingly, due to the simple configuration of the battery pack **200**, it is possible to save the manufacturing cost and effectively reduce the malfunction likelihood.

(49) Referring back to FIGS. 1 to 4, the battery pack **200** of the present disclosure may further include a sensor **262** and a control unit **260**. The sensor **262** may be a gas detection sensor or a temperature sensor configured to sense thermal runaway of the plurality of battery cells **211**. For example, the sensor **262** may be disposed in the at least one cell assembly.

(50) Additionally, when the control unit **260** receives a signal notifying thermal runaway from the sensor **262**, the control unit **260** may be configured to transmit the signal to the switch **225** to electrically connect the cell assembly **210** to the resistor **223**. The control unit **260** may include a communication line **261** to transmit the electrical signal to the switch **225**. For example, when the control unit **260** receives the signal notifying thermal runaway from the sensor **262**, the control unit **260** may be configured to output a control signal for turning on the switch **225**.

(51) According to this configuration of the present disclosure, the present disclosure further includes the control unit **260**, thereby automatically controlling the switch **225** quickly when thermal runaway of the plurality of battery cells **211** is sensed. Accordingly, the present disclosure may quickly consume the power of the cell assembly **210** in which thermal runaway occurred, thereby increasing the safety.

(52) Referring back to FIGS. 6 and 7, when comparing with the energy consumption unit **220** of FIG. 4, the battery pack **200** according to another embodiment of the present disclosure may include the electrical line **221** of a different shape. For example, as shown in FIG. 7, the electrical line **221** of the energy consumption unit **220** may be in the shape of a coil. When the resistor **223** is discharged out of the storage unit **227**, the lengths of the coil shaped electrical line **221** may be elastically extended in the downward direction.

(53) According to this configuration of the present disclosure, the present disclosure includes the coil shaped electrical line **221**, thereby reducing impacts caused by a fall of the resistor **223** using the elastic force of the coil shape when the resistor **223** is discharged out of the storage unit **227**. Accordingly, it is possible to prevent the coil shaped electrical line **221** of the present disclosure from being cut when a force of the falling resistor **223** is transmitted to the electrical line **221**.

(54) Referring back to FIGS. 1 to 4, the battery pack **200** of the present disclosure may include at least two cell assemblies **210**. For example, as shown in FIG. 2, the battery pack **200** may include three cell assemblies **210**.

(55) Additionally, the energy consumption unit **220** may be configured to discharge the cell assembly **210** including the battery cell **211** in which thermal runaway occurred among the at least two cell assemblies **210**.

(56) Further, the energy consumption unit **220** may be configured to discharge the other cell assembly **210** adjacent to the cell assembly **210** including the battery cell **211** in which thermal runaway occurred among the at least two cell assemblies **210**.

(57) According to this configuration of the present disclosure, the present disclosure includes the energy consumption unit **220** configured to discharge the cell assembly **210** including the battery cell **211** in which thermal runaway occurred among the at least two cell assemblies **210**, or configured to discharge the other cell assembly **210** adjacent to the cell assembly **210** including the battery cell **211** in which thermal runaway occurred among the at least two cell assemblies **210**, thereby preventing propagation of thermal runaway from the cell assembly **210** to the adjacent other cell assembly **210** by consuming the power of the cell assembly **210** in which thermal runaway occurred or the cell assembly **210** adjacent to the cell assembly **210** in which thermal runaway occurred without needing to include the energy consumption unit **220** for each of the plurality of cell assemblies **210**.

(58) That is, when thermal runaway occurs in one of the plurality of cell assemblies **210**, propagation of the thermal runaway to the adjacent other cell assembly **210** may be prevented by two methods. One of the methods quickly consumes the power of the cell assembly **210** including the battery cell **211** in which thermal runaway occurred among three cell assemblies **210**, for example, below 30% as shown in FIG. 3. The fast power consumption may reduce the thermal

energy level of the cell assembly **210** in which thermal runaway occurred to a level at which the thermal runaway may not spread to the adjacent other cell assembly **210**, thereby preventing the propagation of the thermal runaway to the other cell assemblies disposed at the front end and the rear end.

(59) Additionally, the other method consumes the power of the front cell assembly **210** and the rear cell assembly **210** adjacent to the cell assembly **210** in which thermal runaway occurred among three cell assemblies. That is, when there are temperature rises of the other cell assemblies **210** disposed at the front end and the rear end by the intermediate cell assembly **210** in which thermal runaway occurred, the power of the other cell assemblies **210** has been already consumed, for example, below 30%, so thermal runaway is less likely to occur, thereby preventing the propagation of the thermal runaway.

(60) FIG. **8** is a schematic perspective view of a battery rack according to an embodiment of the present disclosure. For reference, in FIG. **8**, up, down, left and right directions may be defined when the battery rack is viewed from the F direction.

(61) Referring to FIG. **8**, the battery rack **300** according to an embodiment of the present disclosure may include a rack case **310** that accommodates the plurality of battery packs **200**. The rack case **310** may be configured to receive the plurality of battery packs **200** vertically stacked. In this instance, a part of the energy consumption unit **220** provided in each of the plurality of battery packs **200** may extend out of the rack case **310**. That is, the energy consumption unit **220** may be disposed on the outer side of the battery rack **300** through a part of the rack case **310**.

(62) Further, the rack case **310** may be configured to be open to at least one side (right side). However, the rack case **310** may be configured to close the open side after the energy consumption unit **220** is installed.

(63) Additionally, the battery rack **300** may further include a central control unit **320** including a Battery Management System (BMS) inside or outside of the rack case **310**.

(64) Meanwhile, an energy storage system (not shown) according to an embodiment of the present disclosure may include at least two battery racks **300**. The at least two battery racks **300** may be arranged in a direction. For example, although not shown, the energy storage system may include three battery racks **300** arranged in a direction. The energy storage system may include a central control unit (not shown) to control the charge/discharge of the three battery racks **300**.

(65) The terms indicating directions as used herein such as upper, lower, left, right, front and rear are used for convenience of description only, and it is obvious to those skilled in the art that the term may change depending on the position of the stated element or an observer.

(66) While the present disclosure has been hereinabove described with regard to a limited number of embodiments and drawings, the present disclosure is not limited thereto and it is obvious to those skilled in the art that various modifications and changes may be made thereto within the technical aspects of the present disclosure and the equivalent scope of the appended claims.

DESCRIPTION OF REFERENCE NUMERALS

(67) **200**: Battery pack **250**: Pack housing **210**: Cell assembly **211**: Battery cell **220**: Energy consumption unit **221**, **223**, **225**: Electrical line, Resistor, Switch **227**: Storage unit **227b**, **227b1**: Discharge unit, Discharge door **227c**, **227d**: Stop, Hinge member **240**, **241**, **243**: Cooling unit, Refrigerant, Water tank **260**, **261**, **262**: Control unit, Communication line, Sensor **300**: Battery rack

Claims

1. A battery pack, comprising: at least one cell assembly including a plurality of battery cells; an energy consumer configured to receive power from the at least one cell assembly when thermal runaway occurs in at least one of the plurality of battery cells; and a storage unit in which the energy consumer is received inside, and configured to physically eject at least part of the energy consumer to outside when the energy consumer receives power from the at least one cell assembly.

2. The battery pack according to claim 1, wherein the storage unit includes a discharger configured to support the energy consumer in a direction opposite to a discharge direction of the energy consumer, and wherein the discharger is configured to melt by heat generated by the energy consumer to transfer at least part of the energy consumer to the outside.
 3. The battery pack according to claim 1, wherein the storage unit includes a discharge door which is closed to support in a direction opposite to a discharge direction of the energy consumer, and wherein the discharge door is configured to be released from a closed state and be opened when the thermal runaway occurs.
 4. The battery pack according to claim 1, further comprising: a cooler including a refrigerant to cool the energy consumer ejected to the outside.
 5. The battery pack according to claim 1, wherein the energy consumer includes: a switch; and a resistor which is electrically connected to the at least one cell assembly by operation of the switch and configured to consume electrical energy stored in the at least one cell assembly.
 6. The battery pack according to claim 5, wherein the switch is configured to be turned on by a physical change made by the heat generated from the at least one cell assembly.
 7. The battery pack according to claim 5, further comprising: a sensor to sense the thermal runaway in the plurality of battery cells; and a control unit to output a control signal for turning on the switch to electrically connect the at least one cell assembly to the resistor when the control unit receives a signal notifying the thermal runaway from the sensor.
 8. The battery pack according to claim 5, wherein the at least one cell assembly, the resistor and the switch are interconnected by an electrical line, and wherein the electrical line has a coil shape.
 9. The battery pack according to claim 1, wherein the at least one cell assembly includes at least two cell assemblies, and wherein the energy consumer is configured to receive power from a first cell assembly including the battery cell in which the thermal runaway occurred among the at least two cell assemblies or a second cell assembly among the at least two cell assemblies adjacent to the first cell assembly.
 10. A battery rack, comprising: the battery pack according to claim 1, and a rack case in which the battery pack is received.
 11. An energy storage system comprising at least two battery racks according to claim 10.
 12. The battery pack according to claim 1, wherein the at least one cell assembly includes a first cell assembly, a second cell assembly and a third cell assembly, the second cell assembly being between the first cell assembly and the third cell assembly, and wherein the energy consumer is connected to the second cell assembly.
 13. The battery pack according to claim 12, wherein the energy consumer is connected to the second cell assembly by wires, and wherein the wires include at least one switch.
 14. The battery pack according to claim 13, wherein the energy consumer is connected to a positive terminal and a negative terminal of the second cell assembly.
 15. The battery pack according to claim 14, wherein the at least one switch comprises a first switch between the second cell assembly and the positive terminal and a second switch between the second cell assembly and the negative terminal.
 16. The battery pack according to claim 13, wherein the at least one switch is activated by an elevated temperature.
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